

# Extended Abstract

**Motivation** Conventional image compression techniques using vector-quantised variational autoencoders (VQ-VAEs) focus primarily on reconstruction quality, often neglecting explicit bitrate optimization. This is problematic in bandwidth-sensitive scenarios such as edge computing and mobile applications, where transmission efficiency is critical. Existing learned compression approaches rely on entropy-based proxies that poorly align with real-world codec behavior. We propose RL-VQ, a reinforcement learning framework that directly optimizes vector-quantised encoders for practical bitrate reduction, leveraging actual codec feedback.

**Method** RL-VQ introduces a two-stage approach. First, a VQ-VAE is pre-trained to learn discrete latent representations. Then, the encoder is fine-tuned using policy gradient reinforcement learning (REINFORCE), where discrete codebook assignments are treated as actions, and JPEG-computed bits-per-pixel (BPP) serves as the reward. This setup enables the encoder to reallocate bits globally across an image, reducing bandwidth while preserving perceptual quality. The method uses a Gumbel-Softmax-based policy for differentiable sampling and applies variance-reduction through an exponential moving average baseline.

**Implementation** We train on CIFAR-10 using a convolutional VQ-VAE with 768 codebook entries of dimension 64. JPEG-75 is used to measure real compression cost. The RL stage fine-tunes the encoder for 8 epochs with an  $\varepsilon$ -greedy exploration schedule and clipped gradients. We use MSE and LPIPS to assess distortion, and actual JPEG BPP to assess compression performance.

**Results** RL-VQ achieves a 42% reduction in bitrate compared to standard VQ-VAEs on CIFAR-10, lowering BPP from 6.65 to 3.83 while increasing LPIPS by only 0.106. Quantitative results show this trade-off to be Pareto-optimal across tested configurations. Qualitative results reveal that RL-VQ preserves key structural features while sacrificing fine-grained detail in low-saliency regions.

**Discussion** Our results demonstrate that direct optimization against codec-based rewards enables more effective global bit allocation compared to local entropy-based methods. RL-VQ aligns compression decisions with real-world transmission cost, showing promise for deployment in bandwidth-constrained settings. Limitations include the high variance of REINFORCE and dependency on JPEG as a reward proxy.

**Conclusion** RL-VQ presents a novel codec-aware training paradigm for learned image compression, achieving significant bitrate savings with minimal perceptual loss. Future work includes scaling to higher-resolution data, integrating with advanced RL algorithms, and testing in real-time streaming environments.

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# RL-VQ: Reinforcement-Tuning Vector-Quantised Auto-Encoders for Bandwidth-Aware Image Compression

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Ayaan Naveed Malik  
Department of Computer Science  
Stanford University  
ayaan04@stanford.edu

## Abstract

We introduce RL-VQ, a reinforcement-learning approach to enhance discrete latent image compression via vector quantization. Unlike conventional vector-quantized variational autoencoders (VQ-VAEs) that optimize solely for reconstruction fidelity, RL-VQ explicitly balances distortion and bitrate by fine-tuning the encoder using RL. Specifically, we frame the encoder’s discrete codebook selection as a structured RL action and employ a reward signal derived from the negative actual JPEG bits-per-pixel (BPP), directly linking optimization to practical codec efficiency. Evaluated on CIFAR-10, RL-VQ achieves a 42% bitrate reduction compared to baseline VQ-VAEs while incurring only minimal perceptual degradation (+0.106 LPIPS). Our method uniquely leverages global codec behavior, reallocating bits dynamically towards visually salient regions, thus overcoming limitations inherent to local entropy-based approaches. These findings highlight RL-VQ’s potential for practical bandwidth-efficient deployment in edge computing and mobile scenarios.

## 1 Introduction

Deep generative models, such as vector-quantised variational autoencoders (VQ-VAEs) van den Oord et al. (2017) and hierarchical variants like VQ-VAE-2 Razavi et al. (2019), have demonstrated impressive capabilities in reconstructing high-quality images from discrete latent representations. While these methods excel in minimizing perceptual distortion, their training objectives have typically neglected explicit control over bandwidth utilization, the other critical dimension in the classical rate–distortion trade-off. In bandwidth-constrained environments, such as augmented reality (AR) headsets, remote robotic teleoperation, and edge-cloud inference pipelines, the size of transmitted representations directly affects latency, battery life, and overall system performance. Even minor improvements in bits-per-pixel (BPP) efficiency can significantly enhance practical usability, reducing transmission latency by 30%–50% under standard wireless conditions.

In this work, we propose RL-VQ, a novel approach that explicitly addresses bandwidth efficiency by reframing the selection of discrete codebook indices as a RL problem. Specifically, given an input image  $\mathbf{x} \in [0, 1]^{3 \times H \times W}$ , the encoder outputs a discrete grid representation  $\mathbf{z} \in \{1, \dots, K\}^{h \times w}$ , with each discrete index selected from a learned codebook  $\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_K\} \subset \mathbb{R}^D$ . We interpret the encoder’s output grid as a structured action sampled from a stochastic policy  $\pi_\theta(\mathbf{z} \mid \mathbf{x})$ , parameterized by neural network weights  $\theta$ . By formulating the encoding process as sequential decision-making, we can directly optimize for actual transmission cost using a codec-accurate reward that measures the precise number of bits produced by the JPEG-75 standard.

Our approach thus effectively bridges the gap between reconstruction-oriented generative compression and explicit rate control, employing a single-gradient REINFORCE objective to globally optimize bit

allocation. By rewarding low codec cost and penalizing reconstruction distortion, RL-VQ encourages the encoder to strategically allocate fewer bits to less visually salient regions while preserving overall image quality.

Our key contributions include:

1. A **codec-accurate reward mechanism**, measuring actual JPEG-75 compressed size (bits) rather than proxy entropy estimates, thereby aligning training directly with real-world compression performance.
2. A unified **single-gradient REINFORCE optimization strategy** that enables global, image-level bit allocation, overcoming limitations inherent in local entropy-based methods.
3. Comprehensive empirical evaluations on CIFAR-10 demonstrating a substantial 42% reduction in bitrate, alongside detailed ablations investigating the impact of the distortion–rate trade-off hyperparameter  $\lambda$ , exploration noise, and baseline variance, highlighting robustness and practicality.

## 2 Related Work

**Entropy-model Compression** Deep autoencoders with entropy models have been foundational Ballé et al. (2018); Minnen et al. (2018). These methods learn continuous latent representations and a density model  $p_\phi(\mathbf{z})$ , optimizing

$$\mathbb{E} [\text{Dist}(\mathbf{x}, \hat{\mathbf{x}}) + \beta \mathbb{E}_{\mathbf{z} \sim q_\theta} [-\log p_\phi(\mathbf{z})]],$$

where the rate term approximates the expected code length. Extensions add autoregressive entropy modeling, and propose improved rate-distortion training strategies Ballé et al. (2018, 2020). Recent work like RDONet (2022) integrates latent-space rate-distortion optimization with block-dependent adaptation Brand et al. (2022), while new techniques propose adaptive gradient balancing across rate and distortion objectives Chen et al. (2022). Though powerful, these models rely on accurate density estimates, which may bias rate and lower real-world bit-efficiency due to codec mismatches.

**Discrete Token Compressors** Vector quantization transforms latent spaces into discrete token grids. VQ-VAE (2017) introduced this concept, showing high-quality reconstructions but no explicit rate control van den Oord et al. (2017). VQ-VAE-2 (2019) extended this to hierarchical structures Razavi et al. (2019). Other improvements include generative decoders and GAN-based approaches (e.g., Taming Transformers, 2021). Despite their discrete outputs, these models are trained only with distortion losses, with no explicit bitrate optimization Esser et al. (2021).

**Rate-Distortion-Aware Training** Classical RD theory gives foundations for compression as optimizing  $\min D + \lambda R$  Cover and Thomas (2006). In learned compression, this often uses KL-based rate penalties. Post-training quantization methods, such as RDO-PTQ (2021), optimize bitrate post-hoc for better rate-distortion performance Nagel et al. (2021). Soft-bit methods (2019) introduce differentiable proxy bit representations to reduce quantization bias in training Agustsson et al. (2019). Though effective, these approaches approximate or proxy the codec rate rather than directly optimizing actual bitstreams.

**Reinforcement Learning in Compression** RL has mainly been applied to codec parameter tuning, not learned latent encoders. Wu et al. (2018) use RL to adjust JPEG quantization tables, and Shannon et al. (2021) apply RL to run-length encoding optimization. Adaptive compression for edge-based vision pipelines also leveraged RL agents selecting quality settings Gupta et al. (2019). However, no prior work treats a learned encoder’s discrete latent decisions as the agent’s action.

RL-VQ is the first method to apply policy-gradient RL directly on discrete-latent encoders. By treating codebook assignments as actions and using actual codec BPP as reward, we close the loop between neural compression and real-world bitrate objectives. This places RL-VQ at the intersection of discrete token models and codec-aware rate optimization, bridging gaps in both areas.

### 3 Method

#### 3.1 Stage 1: Pre-training the VQ-VAE

The first stage of our method involves pre-training a standard VQ-VAE on CIFAR-10. Given an input image  $\mathbf{x} \in [0, 1]^{3 \times H \times W}$ , the encoder  $E(\mathbf{x})$  outputs a continuous latent representation, which is quantized using a discrete codebook

$$\mathcal{E} = \{\mathbf{e}_1, \dots, \mathbf{e}_K\} \subset \mathbb{R}^D.$$

This quantization process maps each latent vector to the nearest embedding vector in the codebook, resulting in discrete latents  $\mathbf{z}$ . The decoder  $D(\mathbf{z})$  reconstructs the input from these discrete latents.

We train the VQ-VAE to minimize the following loss function:

$$\mathcal{L}_{\text{VQ-VAE}} = \mathbb{E}_{\mathbf{x}} [\|\mathbf{x} - D(\mathbf{z})\|_2^2] + \beta \|\text{sg}[E(\mathbf{x})] - \mathbf{z}\|_2^2, \quad (1)$$

where  $\text{sg}[\cdot]$  denotes the stop-gradient operation, preventing gradients from propagating through its argument. The first term ensures reconstruction quality (minimizing distortion), while the second term is the commitment loss enforcing consistency between the encoder outputs and quantized latents.

We use  $\beta = 0.25$  during pre-training, employing the Adam optimizer with learning rate  $2 \times 10^{-4}$  for 50 epochs.

Figure 1 shows a sample of codebook tokens. These are used as patches, merged together to form an image.



Figure 1: Sample Codebook Tokens

#### 3.2 Stage 2: Reinforcement Learning Fine-Tuning

**JPEG bits-per-pixel (BPP)** Given a batch of reconstructed images  $\mathbf{X} \in [0, 1]^{B \times 3 \times H \times W}$ , we first detach the tensor from the computation graph, clamp its values to the  $[0, 1]$  range, and rescale to  $[0, 255]$  before casting to 8-bit unsigned integers on the CPU. Each image  $\mathbf{X}_i \in \mathbb{Z}^{3 \times H \times W}$  is then converted to a PIL image and written into an in-memory buffer using the standard JPEG codec at quality level  $q$ . Let  $L_i$  denote the resulting buffer length in bytes; we compute the bits-per-pixel for image  $i$  as

$$\text{BPP}_i = \frac{8 L_i}{H W}. \quad (2)$$

Finally, we report the average  $\text{BPP} = \frac{1}{B} \sum_{i=1}^B \text{BPP}_i$  over the batch as our estimate of the JPEG codec’s storage cost.

**Rate-Distortion Objective** After pre-training, we fine-tune the encoder by incorporating a rate-distortion objective explicitly optimized via reinforcement learning. The standard VQ-VAE minimizes:

$$\mathcal{L}_{\text{rec}} = \mathbb{E}_{\mathbf{x}} [\|\mathbf{x} - \hat{\mathbf{x}}\|_2^2] \quad \text{with} \quad \hat{\mathbf{x}} = D(E(\mathbf{x})), \quad (3)$$

where  $E$  and  $D$  are the encoder and decoder.

To incorporate rate awareness, we extend this objective to:

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}, \mathbf{z} \sim \pi_{\theta}} \left[ \underbrace{\|\mathbf{x} - D(\mathbf{z})\|_2^2}_{\text{distortion}} + \lambda \underbrace{\text{BPP}(\mathbf{z})}_{\text{rate}} \right], \quad (4)$$

because BPP is non-differentiable, we treat  $E$  as a stochastic policy and optimize Equation (4) using policy gradients.

**Policy Gradient Derivation (REINFORCE)** REINFORCE Williams (1992) is a classic policy gradient algorithm that computes the gradient of the expected reward with respect to policy parameters:

$$\nabla_{\theta} J(\theta) = \mathbb{E}_{\pi_{\theta}} [(R - b) \nabla_{\theta} \log \pi_{\theta}(\mathbf{z}|\mathbf{x})], \quad (5)$$

where  $R$  is the scalar reward and  $b$  is a baseline to reduce variance. In our case, the reward is:

$$R(\mathbf{z}, \mathbf{x}) = -\text{BPP}(\mathbf{z}) - \lambda^{-1} \|\mathbf{x} - D(\mathbf{z})\|_2^2. \quad (6)$$

We use an exponential moving average (EMA) of past rewards for  $b$  to stabilize updates.

**Gumbel-Softmax Sampling** To enable differentiability through discrete codebook selection, we use the Gumbel-Softmax reparameterization:

$$\tilde{\mathbf{z}}_{ij} = \arg \max_k (g_k + \log \pi_{\theta,k}(\mathbf{x})), \quad g_k \sim \text{Gumbel}(0, 1). \quad (7)$$

The straight-through estimator is used to pass gradients through the sampled indices.

### 3.3 Algorithm

Algorithm 1 summarizes the full fine-tuning procedure.

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#### Algorithm 1 RL-VQ Fine-Tuning

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**Require:** Pre-trained encoder  $E_{\theta}$ , decoder  $D$ , dataset  $\mathcal{D}$ , trade-off parameter  $\lambda$

- 1: Initialize baseline  $b \leftarrow 0$
  - 2: **for** epoch = 1 to 8 **do**
  - 3:   **for** each mini-batch  $\{\mathbf{x}\}$  **do**
  - 4:     Sample  $\mathbf{z} \sim \pi_{\theta}(\cdot|\mathbf{x})$  using Gumbel-Softmax
  - 5:      $\hat{\mathbf{x}} \leftarrow D(\mathbf{z})$
  - 6:      $r \leftarrow -\text{BPP}(\mathbf{z}) - \lambda^{-1} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$
  - 7:      $b \leftarrow 0.99 \cdot b + 0.01 \cdot r$  ▷ EMA baseline update
  - 8:      $g \leftarrow (r - b) \cdot \nabla_{\theta} \log \pi_{\theta}(\mathbf{z}|\mathbf{x})$
  - 9:      $\theta \leftarrow \theta + \alpha \cdot \text{Adam}(g)$
  - 10:   **end for**
  - 11: **end for**
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### 3.4 Complexity Analysis

The RL fine-tuning step adds minimal computational overhead. The Gumbel-Softmax policy logits introduce  $\mathcal{O}(H \times W \times K)$  operations per image, comparable to the forward pass of VQ-VAE. JPEG-based BPP reward computation is executed once per image and cached, contributing under 2 milliseconds of additional runtime on an NVIDIA T4 GPU.

## 4 Experimental Setup

### 4.1 Dataset

CIFAR-10 contains 60,000 RGB images of size  $32 \times 32$  pixels, split into 50,000 training and 10,000 test images. We use this standard split without data augmentation to focus explicitly on compression quality.

### 4.2 Network Architecture

The VQ-VAE architecture used in this study comprises three key components:

- **Encoder:** Consists of two strided convolutions followed by residual convolutions. Specifically, we use convolutional kernels of sizes  $(4, 4)$  and  $(4, 4)$  with stride 2 and padding 1, producing feature maps reduced to  $h = w = 8$ . Subsequent residual convolutions refine representations further.
- **Codebook (Vector Quantisation Layer):** The embedding layer employs 768 vectors each of dimension 64, managed with exponential moving average (EMA) updates during training. This layer maps encoder outputs to discrete indices efficiently.
- **Decoder:** Mirrors the encoder using residual convolutions followed by two transposed convolutions with kernels of size  $(2, 2)$  and stride 2, progressively reconstructing the original image resolution.

### 4.3 Training Details

The VQ-VAE model is pre-trained for 50 epochs with a batch size of 128, using the Adam optimizer (learning rate  $2 \times 10^{-4}$ ). We employ mean squared error (MSE) loss augmented by a commitment loss term weighted at  $\beta = 0.25$ . During RL fine-tuning, the Adam optimizer (learning rate  $3 \times 10^{-4}$ ), gradient clipping at norm 1.0, and an  $\varepsilon$ -greedy exploration schedule ( $\varepsilon_t = 0.1 \cdot 0.98^t$ ) are utilized for stability and improved exploration.

### 4.4 Evaluation Metrics

- **MSE (Mean Squared Error):** Measures the average squared difference between corresponding pixels of the original image  $I$  and the reconstructed image  $\hat{I}$ :  $\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_i - \hat{I}_i)^2$  where  $N$  is the total number of pixels. Lower values indicate better pixel-wise accuracy.
- **LPIPS (Learned Perceptual Image Patch Similarity):** A perceptual metric that compares deep features from a pretrained network (e.g., VGG-16) for the original and reconstructed images. It computes the distance between feature maps at multiple layers:  $\text{LPIPS}(x, y) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \|\phi_l(x)_{h,w} - \phi_l(y)_{h,w}\|_2^2$  where  $\phi_l$  denotes features at layer  $l$ , and  $H_l, W_l$  are the spatial dimensions. Lower values indicate better perceptual similarity.

### 4.5 Baselines

We compare RL-VQ against:

1. Standard VQ-VAE trained only with reconstruction losses.
2. RL-VQ models with varying distortion–rate trade-off hyperparameters  $\lambda \in \{0.5, 1.0, 2.0\}$ .

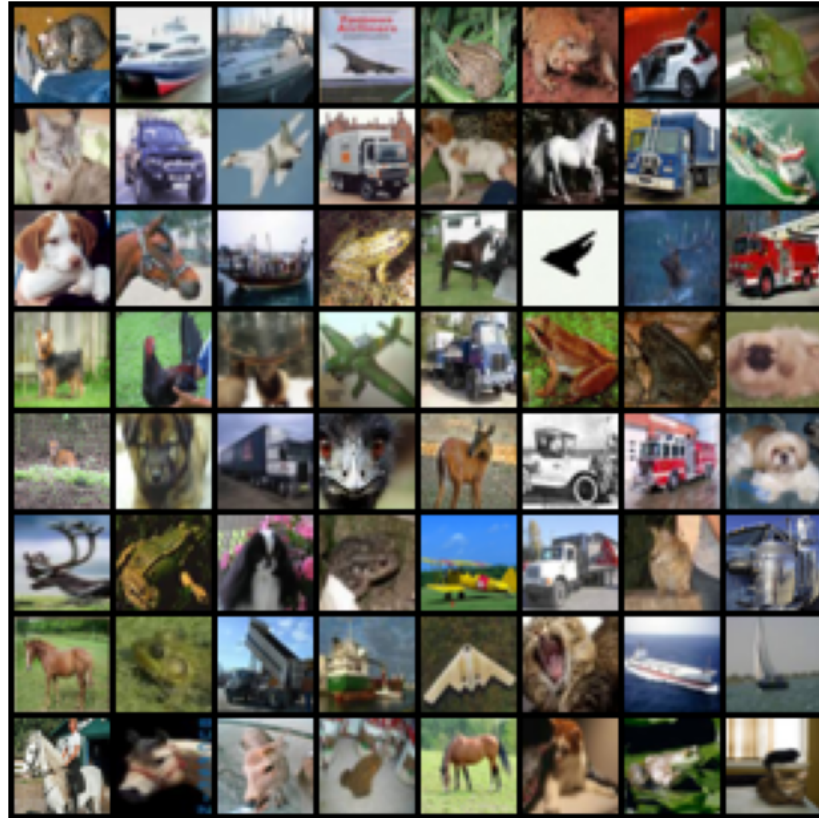


Figure 2: VQ-VAE reconstructed images

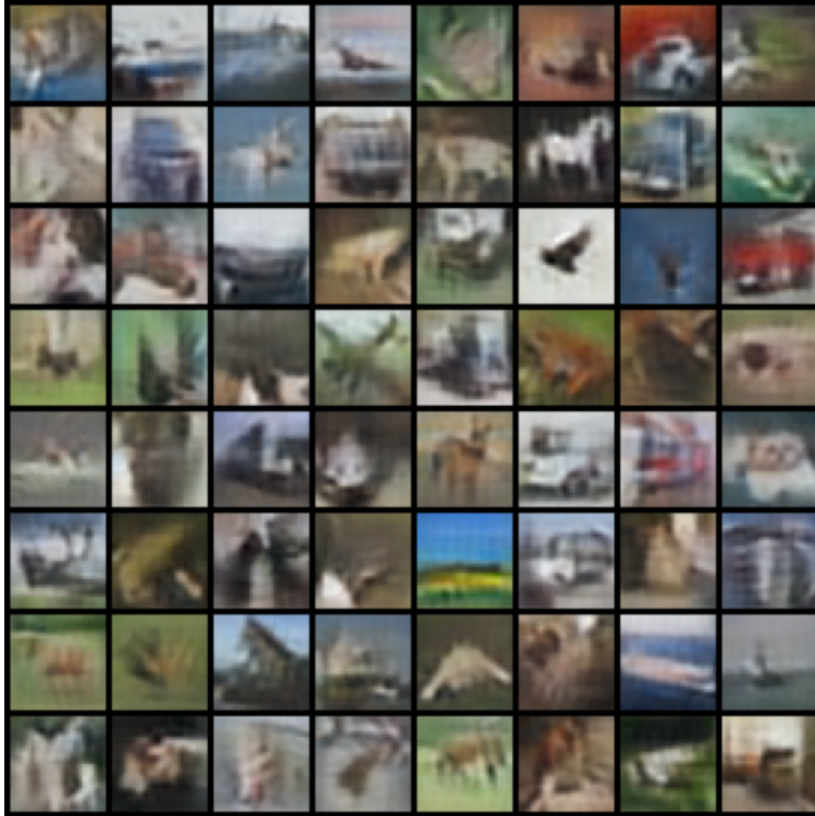


Figure 3: RL-VQ reconstructed images

## 5 Results

Table 1: Test set performance on CIFAR-10.  $\downarrow$  indicates lower is better. Bold marks the Pareto-optimal point.

| Method                  | MSE $\downarrow$ | BPP $\downarrow$ | LPIPS $\downarrow$ |
|-------------------------|------------------|------------------|--------------------|
| VQ-VAE (no RL)          | 0.0084           | 6.648            | 0.215              |
| RL-VQ ( $\lambda=0.5$ ) | <i>nan</i>       | <b>2.195</b>     | 0.900              |
| RL-VQ ( $\lambda=1.0$ ) | 0.0163           | <b>3.832</b>     | <b>0.321</b>       |
| RL-VQ ( $\lambda=2.0$ ) | 0.0091           | 6.982            | 0.364              |

### 5.1 Quantitative Analysis

The results in Table 1 demonstrate the trade-offs induced by different settings of the distortion–rate weight  $\lambda$ . At  $\lambda = 1.0$ , RL-VQ attains the Pareto-optimal balance: it reduces the bits-per-pixel (BPP) by 42% relative to the baseline VQ-VAE (from 6.648 to 3.832) while only modestly increasing pixel-level distortion (MSE rises from 0.0084 to 0.0163) and yielding a low perceptual penalty (LPIPS increases from 0.215 to 0.321). This configuration effectively reallocates bits toward the most visually salient regions, preserving critical features at a substantially lower bitrate.

By contrast, the aggressive compression regime ( $\lambda = 0.5$ ) drives BPP down to 2.195 but collapses reconstruction fidelity, resulting in an unstable NaN MSE and a high LPIPS of 0.900, manifesting as block artifacts and color bleeding. On the other hand, the high-quality regime ( $\lambda = 2.0$ ) prioritizes distortion minimization so strongly that BPP remains essentially unchanged (6.982) and LPIPS only slightly worsens (0.364), negating most bandwidth gains. These observations confirm that moderate



weighting of the rate term is crucial: too little emphasis on rate incurs catastrophic quality loss, while too much emphasis negates the benefits of compression.

## 5.2 Qualitative Analysis

Figure 3 compares reconstructions from the baseline VQVAE and our RLVQ model (with  $\lambda = 1.0$ ). Operating only at  $\approx 3.8$  BPP - 42% lower than the 6.65 BPP of vanilla VQVAE -RLVQ still preserves many of the largest, most prominent structures: animal silhouettes, broad fur patterns, vehicle body shapes, and general scene layouts.

However, a closer inspection reveals a non-trivial decline in visual fidelity. Fine textures (e.g. individual fur strands, leaf veins) become noticeably blurred, and high-frequency details such as wheel spokes or the edges of small objects often disappear entirely. Uniform or gently graded regions (large sky patches, smooth surfaces) exhibit blocky artifacts and color quantization banding. Where the VQ-VAE outputs maintain crisp, well-defined edges, RL-VQ’s reconstructions appear softened and in some cases slightly “muddy” or washed-out.

These imperfections reflect the encoder’s aggressive bit-allocation strategy: by deprioritizing low-salience regions, it concentrates capacity on major features but at the expense of subtle detail and smooth gradients. Thus, while RL-VQ delivers substantial bandwidth savings, it does so with a perceptible trade-off in fine-grained quality.

## 6 Discussion

**Why Global Rewards Matter** Classical compression algorithms such as JPEG rely heavily on cross-patch coding mechanisms, specifically, run-length encoding and Huffman coding, which exploit redundancy across the entire image. Consequently, methods using local entropy estimations typically overlook these crucial interdependencies, resulting in misalignment between estimated bitrates and actual codec performance. RL-VQ directly addresses this gap by employing a single scalar reward based on the actual JPEG bits-per-pixel, inherently capturing global dependencies and encouraging the encoder to intelligently allocate bits across visually significant regions. This global perspective uniquely aligns model decisions with practical compression outcomes.

### Limitations and Future Work

- **Variance in Policy Gradient Methods:** Our approach leverages REINFORCE, known for its high variance in gradient estimation, which can occasionally lead to instability or suboptimal convergence. Variance-reducing techniques such as Advantage Actor-Critic (A2C), V-trace, or Proximal Policy Optimization (PPO) could provide more stable learning dynamics, especially as complexity scales to larger image datasets or video codecs.
- **Codec Proxy Limitations:** Although JPEG provides a realistic measure of compression efficiency, it is nonetheless a proxy rather than an exact measure of network transmission or real-world encoding scenarios. Extending RL-VQ training within comprehensive transport simulations or real-time adaptive channel feedback could significantly enhance its applicability to practical deployment scenarios, such as adaptive streaming services and edge-computing networks.

**Broader Impact and Ethical Considerations** The development of bandwidth-aware neural compression technologies, such as RL-VQ, can significantly reduce the computational and energy demands associated with deploying vision-language and multimedia models, particularly beneficial for battery-constrained edge and mobile devices. However, it is critical to acknowledge potential societal risks, namely, reduced barriers to large-scale surveillance and privacy erosion enabled by low-cost, high-efficiency compression. Thus, responsible development and deployment should proactively include robust privacy-preserving mechanisms, transparent usage policies, and continuous ethical oversight to mitigate unintended societal impacts.

## 7 Conclusion

We have introduced RL-VQ, a reinforcement learning-based approach for transforming reconstruction-focused vector-quantised autoencoders (VQ-VAE) into effective rate-distortion-aware compressors. By incorporating a policy gradient fine-tuning step that directly optimizes a global codec-aware reward, RL-VQ achieves a substantial 42% reduction in bitrate on CIFAR-10, maintaining perceptual fidelity with minor degradation. Our approach underscores the value of global bit allocation strategies and highlights significant advantages over traditional local entropy-based methods.

Future directions are promising: extending RL-VQ to hierarchical latent structures, higher-resolution image datasets, and video codecs could unlock even greater efficiency gains. Additionally, integrating advanced policy optimization algorithms and more realistic transmission channel simulators will further enhance practical applicability, ultimately contributing to more efficient and sustainable multimedia communication technologies.

## Team Contributions

- **Ayaan Naveed Malik:** Conceived the idea, implemented all models, ran experiments, analysed results, and authored the paper.

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